TrES-2b

R-0064

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-2_210414` · created 2026-07-15T15:22:14+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459318.86788 +/- 0.00077 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.114 +/- 0.021
Transit depth (Rp/Rs)^2	1.3 +/- 0.48 [%]
Orbital Inclination (inc)	83.82 +/- 1.18
Airmass coefficient 1 (a1)	1.045 +/- 0.0096
Airmass coefficient 2 (a2)	-0.0253 +/- 0.0034
Scatter in the residuals of the lightcurve fit is	0.91 %
Best Comparison Star	#3 - [374, 140]
Optimal Aperture	4.09
Optimal Annulus	15.85
Transit Duration (day)	0.066 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.114 ± 0.021 vs 0.1254 (deviation 0.52σ)
Duration fit vs published	0.066 d vs 0.0746 d (deviation 0.34σ)
Mid-time O–C	1.9 min
Depth SNR	5.2
Residual scatter	0.91 %

Light curve

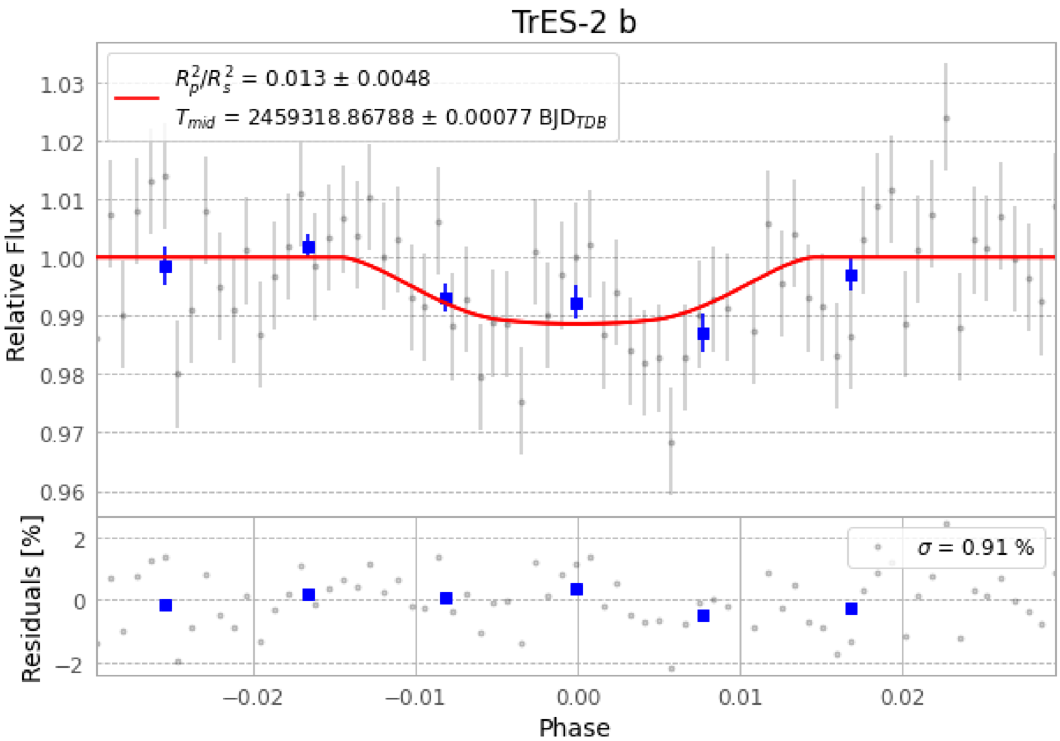

Figure 1 — FinalLightCurve_TrES-2 b_2021-04-14.png (SHA-256 2e6d35d7ca3fcbd0…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [276, 302], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 774c0fe28e3742f7…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

71 FITS frames (47 MB), `TRES-2210414070312.FITS` → `TRES-2210414103312.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-210414.log (8b28380ec2b5…), FinalLightCurve_TrES-2 b_2021-04-14.png (2e6d35d7ca3f…), FinalLightCurve_TrES-2 b_2021-04-14.csv (5e8ed6ee901a…)

Compute resources

Wall time 125.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)